Xinyu (Catherine) Tian

+1 619-380-4826 · xit026@ucsd.edu
bci-plumber.github.io · github.com/bci-plumber

EDUCATION

Jul 2026 – Jan 2027

University of California San Diego La Jolla, CA
Visiting Student · Swartz Center for Computational Neuroscience, Jung Lab

Sep 2023 – Jun 2027

Tsinghua University Beijing, China
B.S. in Biology, Zhili College · GPA 3.9 / 4.0 · rank 3 / 35
– Advisors: Prof. Xiaorong Gao, Prof. Bo Hong

PUBLICATIONS

2026

GazeAgent: VLM-Guided Implicit Gaze Calibration from Egocentric Multimodal Behavior for AI Glasses
*Yuhong Zhang, Khanh An Nguyen, **Xinyu Tian**, Yuchen Xu, KungYi Chang, Abraham Padua, Ying Wu, Gert Cauwenberghs, Tzyy-Ping Jung · Under review at ACM CHI*

RESEARCH

Egocentric Gaze and Vision–Language Models

Apr – Sep 2026

Deriving Pseudo-Calibration Targets from Precise Visual Attention
UC San Diego, Jung Lab (remote Apr – Jun) · Part of GazeAgent (CHI submission, third author)
– Benchmarked nine VLMs on validating precise-visual-attention events and localizing their targets from egocentric video, gaze and IMU streams
– Fine-tuned Qwen3.5-35B-A3B with LoRA on human-consensus labels, raising validation accuracy from 75.0% to 89.5% (Claude Opus 4.6: 81.4%)
– Designed the recall-oriented multimodal detector and the annotation criteria; five annotators converged to a median of 1.93° on targets of median width 13°

Apr – Jun 2026

Is Egocentric Gaze Autoregressive? Toward a Gaze Foundation Model
Independent project (remote)
– Built a gaze-tokenization pipeline on Microsoft HoloAssist: projected HoloLens gaze into RGB frames via depth (758k samples, 143 sessions), with five gaze representations and eight tokenizers feeding transformer sequence models
– Found next-fixation prediction dominated by temporal continuity: the best model beat a persistence baseline by only 3.2%, and instruction text added no gain, suggesting gaze needs external task context rather than autoregression
– This finding informed the context-driven design of GazeAgent above

Jul 2026 – present

Causal Alignment of Human Gaze and Vision–Language Model Attention
UC San Diego · GROUP LEADER
– Tested whether human fixation causally drives model attention in LLaVA and Qwen2-VL, patching activations at gaze-peak tokens against ground-truth fixations from the VQA-MHUG corpus
– Ablating gaze-peak tokens produced a significantly larger effect than matched random ablation (preliminary; $p < 0.01$, $N = 120$)
– Implemented in TransformerLens, following the ARENA 3.0 mechanistic-interpretability methodology

Brain–Computer Interfaces for Real-Time Control

Feb 2025 – Jan 2026

Low-Fatigue Stimulus Design for SSVEP Spellers
Tsinghua University, Gao Lab · GROUP LEADER
– Replaced high-depth brightness flicker with low-depth and text-based stimuli, targeting the flicker sensation that limits how long a speller can be used rather than its peak accuracy
– Encoded three levels of Chinese linguistic structure — zi (character), ci (word) and chengyu (phrase) — to improve decoding robustness across units of different length
– Benchmarked FBCCA, TDCA and TRCA on accuracy and runtime across EEG from ten subjects; the low-depth study was conducted independently

Jun – Sep 2025

Real-Time Wheelchair Navigation via a Text-Based SSVEP Interface

Tsinghua University, Gao Lab · GROUP MEMBER

- Built a closed-loop interface issuing seven directional commands — forward, back, left, right, two rotations and stop — at 0.5 s decision intervals using calibrated TDCA
- Presented text-coded stimuli through an AR headset, trading peak decoding accuracy for stimuli a user can tolerate across a full session
- Validated and debugged the deployed system for the World Robot Contest

Computer Vision and Signal Processing for Clinical Assessment

Jan – Mar 2026

Automatic Detection of SPES Artifacts in ECoG

King's College London · GROUP LEADER

- Built an EEGLAB plugin that flags single-pulse electrical stimulation artifacts for seizure-onset-zone identification, replacing exhaustive visual inspection with a human-in-the-loop review step
- Across the five clinical recordings available, tuned detection caught every artifact with under 5% false positives and cut review time by roughly 80% (approximate figures on a small sample)
- Deliberately rule-based rather than learned, so a clinician can predict and audit its behaviour

Aug 2025 – present

Stereo Vision-Based ARAT Automation as Paralysis Rehabilitation Benchmark

Tsinghua University, Hong Lab · GROUP LEADER

- Designed the system architecture for scoring the Action Research Arm Test from stereo camera video, combining binocular stereo vision with CAD-based 6D pose estimation; implementation is ongoing
- Proposed tensor decomposition for dimensionality reduction in motor-imagery decoding
- Part of EBR, a minimally invasive BCI rehabilitation system funded as a key project by the Tsinghua Innovative Research Commission; supported ECoG data collection from patients

COURSEWORK

GRADED Linear Algebra · Calculus · Probability & Statistics · Fundamentals of Computer Programming (Python) · Biostatistics · Data Science in Microbiology

SKILLS

INTERPRETABILITY TransformerLens · activation patching · ablation and causal intervention

MACHINE LEARNING Python · PyTorch · LoRA fine-tuning of LLMs / VLMs · multi-model benchmarking

SIGNAL PROCESSING EEG & ECoG (FBCCA, TDCA, TRCA) · tensor decomposition · MATLAB / EEGLAB

LANGUAGES English (IELTS 8.0) · Mandarin (native) · Arabic (elementary)

TALKS

Jun 2026

Active Use of Latent Tree-Structured Sentence Representation in Humans and Large Language Models

Journal club · Gao Lab Meeting

Nov 2025

NetFormer: An Interpretable Model for Recovering Dynamical Connectivity in Neuronal Population Dynamics

Journal club · Hong Lab Meeting

Oct 2025

Dream Engineering: From Targeted Memory Reactivation to Visual Reconstruction

Symposium · Tsinghua Academy Elite Scholars Program

Aug 2025

From Psychoacoustic Category Boundaries to Cortical Feature-Tuned Ensembles: Edward Chang's Research on Speech Perception

Journal club · Hong Lab Meeting

Mar 2025

Boolean Network Models: Toward Interpretable and Robust Biomodeling

Symposium · Tsinghua Academy Elite Scholars Program · with Zezhao Wu